

Martin-Luther-Universität Halle-Wittenberg
Faculty of Natural Science II
Institute of Physics

Physics Modelling Assistant

Using Agentic Artificial Intelligence in physics
modelling

Bachelor's Thesis
for the Award of the Academic Degree
Bachelor of Science
in Physik und Digitale Technologien

submitted by: **Janis Felix Jacobi**
Matriculation number: 223308195

Supervisor 1: Prof. Dr. Niels Schröter
Supervisor 2: Prof. Dr. Samir Lounis

submission: Halle (Saale), 23rd August 2026

Contents

1 title 4

1.1 dsf 4

2 Appendix 8

3 Acknowledgement 9

4 Declaration of authorship 10

Figure Index

testFig	4
-------------------	---

Table of Tables

Detector parameters	4
-------------------------------	---

Figure 1: testC

Table 1: Detector parameters

a	b
c	d

1 title

1.1 dsf

References

- [1] Satoshi Ando, Yukio Tanaka, Mario Cuoco, Luca Chirolli, and Maria Teresa Mercaldo. “Spin and orbital Edelstein effect in spin-orbit coupled noncentrosymmetric superconductors”. In: *Physical Review B* 110.18 (11/2024). ISSN: 2469-9969. DOI: [10.1103/physrevb.110.184503](https://doi.org/10.1103/physrevb.110.184503). URL: <http://dx.doi.org/10.1103/PhysRevB.110.184503>.
- [2] arXiv. *arXiv API Access*. 07/09/2026. URL: <https://info.arxiv.org/help/api/index.html>.
- [3] Ajay Bandi, Bhavani Kongari, Roshini Naguru, Sahitya Pasnoor, and Sri Vidya Vilipala. “The Rise of Agentic AI: A Review of Definitions, Frameworks, Architectures, Applications, Evaluation Metrics, and Challenges”. In: *Future Internet* 17.9 (2025). ISSN: 1999-5903. DOI: [10.3390/fi17090404](https://doi.org/10.3390/fi17090404). URL: <https://www.mdpi.com/1999-5903/17/9/404>.
- [4] Sébastien Bubeck, Christian Coester, Ronen Eldan, et al. *Early science acceleration experiments with GPT-5*. 2025. arXiv: [2511.16072](https://arxiv.org/abs/2511.16072) [cs.CL]. URL: <https://arxiv.org/abs/2511.16072>.
- [5] crewai. *core.py*. 04/22/2026. URL: <https://github.com/crewAIInc/crewAI/blob/main/lib/crewai/src/crewai/agent/core.py> (visited on 04/22/2026).
- [6] crewai. *crew.py*. 04/22/2026. URL: <https://github.com/crewAIInc/crewAI/blob/main/lib/crewai/src/crewai/crew.py> (visited on 04/22/2026).
- [7] crewai. *crewAi*. 04/22/2026. URL: <https://crewai.com/> (visited on 04/22/2026).
- [8] crewai. *task.py*. 04/22/2026. URL: <https://github.com/crewAIInc/crewAI/blob/main/lib/crewai/src/crewai/task.py> (visited on 04/22/2026).
- [9] critPt. *challenges*. 04/22/2026. URL: https://github.com/CritPt-Benchmark/CritPt/tree/main/data/public_test_challenges (visited on 04/22/2026).
- [10] Gesellschaft für wissenschaftliche Datenverarbeitung mbH Göttingen. *AcademicCloud*. 04/21/2026. URL: <https://id.academiccloud.de> (visited on 04/22/2026).
- [11] Gesellschaft für wissenschaftliche Datenverarbeitung mbH Göttingen. *SAIA*. 04/21/2026. URL: <https://docs.hpc.gwdg.de/services/ai-services/saia/index.html> (visited on 04/22/2026).
- [12] Hugging Face. *Devstral 2 123B Instruct 2512*. 04/22/2026. URL: <https://huggingface.co/mistralai/Devstral-2-123B-Instruct-2512> (visited on 04/22/2026).
- [13] Irene Gaiardoni, Mattia Trama, Alfonso Maiellaro, Claudio Guarcello, Francesco Romeo, and Roberta Citro. “Edelstein Effect in Isotropic and Anisotropic Rashba Models”. In: *Condensed Matter* 10.1 (03/2025), p. 15. ISSN: 2410-3896. DOI: [10.3390/condmat10010015](https://doi.org/10.3390/condmat10010015). URL: <http://dx.doi.org/10.3390/condmat10010015>.
- [14] Daya Guo, Dejian Yang, Haowei Zhang, et al. “DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning”. In: *Nature* 645.8081 (2025), pp. 633–638. ISSN: 1476-4687. DOI: [10.1038/s41586-025-09422-z](https://doi.org/10.1038/s41586-025-09422-z). URL: <http://dx.doi.org/10.1038/s41586-025-09422-z>.
- [15] Shuai Guo. *LangGraph 101: Let’s Build A Deep Research Agent*. en-US. 08/2025. URL: <https://towardsdatascience.com/langgraph-101-lets-build-a-deep-research-agent/> (visited on 03/12/2026).

- [16] InsightPartners. *CrewAI Launches Multi-Agent Platform to Deliver on the Promise of Generative AI for Enterprise*. 10/23/2023. URL: <https://www.insightpartners.com/ideas/crewai-launches-multi-agent-platform-to-deliver-on-the-promise-of-generative-ai-for-enterprise/> (visited on 04/22/2026).
- [17] Geoffrey Irving, Paul Christiano, and Dario Amodei. *AI safety via debate*. 2018. arXiv: 1805.00899 [stat.ML]. URL: <https://arxiv.org/abs/1805.00899>.
- [18] Annika Johansson. “Theory of spin and orbital Edelstein effects”. en. In: *Journal of Physics: Condensed Matter* 36.42 (07/2024). edelstein effect, p. 423002. ISSN: 0953-8984. DOI: 10.1088/1361-648X/ad5e2b. URL: <https://doi.org/10.1088/1361-648X/ad5e2b> (visited on 03/06/2026).
- [19] Annika Johansson, Borge Göbel, Jürgen Henk, Manuel Bibes, and Ingrid Mertig. “Spin and orbital Edelstein effects in a two-dimensional electron gas: Theory and application to SrTiO₃ interfaces”. In: *Phys. Rev. Res.* 3 (1 03/2021), p. 013275. DOI: 10.1103/PhysRevResearch.3.013275. URL: <https://link.aps.org/doi/10.1103/PhysRevResearch.3.013275>.
- [20] Sergio Leiva-Montecinos, Jürgen Henk, Ingrid Mertig, and Annika Johansson. “Spin and orbital Edelstein effect in a bilayer system with Rashba interaction”. In: *Physical Review Research* 5.4 (12/2023). ISSN: 2643-1564. DOI: 10.1103/physrevresearch.5.043294. URL: <http://dx.doi.org/10.1103/PhysRevResearch.5.043294>.
- [21] Aman Madaan, Niket Tandon, Prakhar Gupta, et al. *Self-Refine: Iterative Refinement with Self-Feedback*. 2023. arXiv: 2303.17651 [cs.CL]. URL: <https://arxiv.org/abs/2303.17651>.
- [22] Agada Joseph Oche and Arpan Biswas. *Role of Large Language Models and Retrieval-Augmented Generation for Accelerating Crystalline Material Discovery: A Systematic Review*. 2025. arXiv: 2508.06691 [cond-mat.mtrl-sci]. URL: <https://arxiv.org/abs/2508.06691>.
- [23] *Open Deep Research*. en. 07/2025. URL: <https://blog.langchain.com/open-deep-research/> (visited on 03/12/2026).
- [24] Stuart Russell and Peter Norvig. *Künstliche Intelligenz Ein moderner Ansatz*. Pearson Benelux B.V. (Zweigniederlassung Deutschland), 2023. ISBN: 9783868944303. URL: <https://elibrary.pearson.de/book/99.150005/9783863263263>.
- [25] Qwen Team. *Qwen3.5-Omni Technical Report*. 2026. arXiv: 2604.15804 [cs.CL]. URL: <https://arxiv.org/abs/2604.15804>.
- [26] SymPy Development Team. *Welcome to SymPy’s documentation!* 04/27/2025. URL: <https://docs.sympy.org/latest/index.html> (visited on 07/09/2026).
- [27] Arunkumar V, Gangadharan G. R., and Rajkumar Buyya. *Agentic Artificial Intelligence (AI): Architectures, Taxonomies, and Evaluation of Large Language Model Agents*. 2026. arXiv: 2601.12560 [cs.AI]. URL: <https://arxiv.org/abs/2601.12560>.
- [28] Dixuan Wang, Yanda Li, Junyuan Jiang, Zepeng Ding, Ziqin Luo, Guochao Jiang, Jiaqing Liang, and Deqing Yang. *Tokenization Matters! Degrading Large Language Models through Challenging Their Tokenization*. 2025. arXiv: 2405.17067 [cs.CL]. URL: <https://arxiv.org/abs/2405.17067>.
- [29] Xiaoxuan Wang, Ziniu Hu, Pan Lu, Yanqiao Zhu, Jieyu Zhang, Satyen Subramaniam, Arjun R. Loomba, Shichang Zhang, Yizhou Sun, and Wei Wang. “SciBench: Evaluating College-Level Scientific Problem-Solving Abilities of Large Language Models”. In: *Proceedings of the Forty-First International Conference on Machine Learning*. 2024.

- [30] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. *Self-Consistency Improves Chain of Thought Reasoning in Language Models*. 2023. arXiv: 2203.11171 [cs.CL]. URL: <https://arxiv.org/abs/2203.11171>.
- [31] Yanzhen Wang, Yiyang Jiang, Diana Golovanova, et al. *QMBench: A Research Level Benchmark for Quantum Materials Research*. 2025. arXiv: 2512.19753 [cond-mat.mtrl-sci]. URL: <https://arxiv.org/abs/2512.19753>.
- [32] Xueqing Xu, Boris Bolliet, Adrian Dimitrov, Andrew Laverick, Francisco Villaescusa-Navarro, Licong Xu, and Íñigo Zubeldia. *Evaluating Retrieval-Augmented Generation Agents for Autonomous Scientific Discovery in Astrophysics*. 2025. arXiv: 2507.07155 [astro-ph.IM]. URL: <https://arxiv.org/abs/2507.07155>.
- [33] Xinyu Zhang, Yuxuan Dong, Yanrui Wu, Jiaying Huang, Chengyou Jia, Basura Fernando, Mike Zheng Shou, Lingling Zhang, and Jun Liu. *PhysReason: A Comprehensive Benchmark towards Physics-Based Reasoning*. 2025. arXiv: 2502.12054 [cs.AI]. URL: <https://arxiv.org/abs/2502.12054>.
- [34] Wayne Xin Zhao, Kun Zhou, Junyi Li, et al. *A Survey of Large Language Models*. 2026. arXiv: 2303.18223 [cs.CL]. URL: <https://arxiv.org/abs/2303.18223>.
- [35] Minhui Zhu, Minyang Tian, Xiaocheng Yang, et al. *Probing the Critical Point (CritPt) of AI Reasoning: a Frontier Physics Research Benchmark*. 2025. arXiv: 2509.26574 [cs.AI]. URL: <https://arxiv.org/abs/2509.26574>.
- [36] Yoel Zimmermann, Adib Bazgir, Alexander Al-Feghali, et al. “32 examples of LLM applications in materials science and chemistry: towards automation, assistants, agents, and accelerated scientific discovery”. en. In: *Machine Learning: Science and Technology* 6.3 (09/2025), p. 030701. ISSN: 2632-2153. DOI: 10.1088/2632-2153/ae011a. URL: <https://doi.org/10.1088/2632-2153/ae011a> (visited on 03/12/2026).

2 Appendix

3 Acknowledgement

4 Declaration of authorship

I hereby declare that I have written this thesis independently and without unauthorized assistance.

All sources and references used have been properly cited.

This thesis has not been submitted in the same or substantially similar form for any other degree or academic qualification.

Halle (Saale), 23rd August 2026

Janis Felix Jacobi